

SHAIK NAFEESA KHAISAR

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SKILLS

Programming Languages: C, C++, Python

Tools & Platforms: Git, Linux, Docker, Arduino IDE, STM32CubeIDE

Technical Skills: Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, NLP, LLM, Retrieval-Augmented Generation, LangChain, LangGraph, LangSmith, PyTorch, MLflow, DVC, AWS SageMaker, Hugging Face, Apache Airflow, ETL Pipelines, Feature Engineering, Model Optimization, FAISS, Pinecone, Weaviate, CAN, UDS, MQTT, DBus, Embedded C (MISRA C)

Hardware & Embedded: STM32, ESP32, Arduino

Soft Skills: Problem Solving, Teamwork, Communication

WORK EXPERIENCE

TVS Motor Company, Bengaluru

July 2024 – Present

Software Engineer (R&D)

RAG-based UDS Service Tool Automation

- Engineered an advanced **RAG system** using **LangChain/LangGraph** to automate **ISO 14229 (UDS)** command generation, cutting manual lookup effort by **50%**.
- Built **ETL pipelines** to vectorize diagnostic data from **CAN logs, DBC, and CDD** files into a **FAISS store**.
- Deployed **LangSmith reasoning agents** to map natural language queries to precise **SIDs and DIDs**.
- Designed **SOA modules** to format raw **hexadecimal messages** for seamless client-to-ECU communication.
- Validated protocol compliance on **CAN networks** using **Vector CANalyzer** for **100% diagnostic accuracy**.

Intelligent Telematics & CAN Anomaly Detector

- Developed a **PyTorch LSTM pipeline** to detect **CAN bus** message starvation and congestion in connected vehicles.
- Performed **Feature Engineering** on **CAN signals** to establish timing and arbitration priority baselines.
- Utilized **Vector CANalyzer** to validate model performance against real-world **CAN matrix** payloads.
- Deployed **MQTT (C++)** clients for real-time **telemetry streaming** from the **TCU** to cloud systems.
- Implemented **MISRA C compliant logic**, improving reliability by **90%** through proactive **anomaly detection**.

EDUCATION

BVRIT Hyderabad College of Engineering for Women

2020 – 2024

Bachelor of Technology in Electrical and Electronics Engineering

CGPA: 7.60 / 10.0

Sri Nalanda Junior College

2018 – 2020

Intermediate (Mathematics, Physics, Chemistry)

898 / 1000

Sri Chaitanya School

2018

Class 10

CGPA: 9.5 / 10

PROJECTS

Battery Cycle Life Prediction using Machine Learning

Python, MATLAB, Machine Learning

- Extracted early-cycle features and trained Machine Learning models achieving ~90% prediction accuracy.
- Published Paper: “**Prediction of Battery Life Cycle Using Machine Learning Algorithm**” at ICITSS 2024.

IIIT Hyderabad — Plug Load Monitoring (IoT)

IoT, ThingSpeak, OM2M, Postman

- Developed an **IoT-based monitoring system** for real-time tracking and built a data collection pipeline using **OM2M**.

ACHIEVEMENTS & CERTIFICATIONS

- Student Innovation Mentor** — Mentored IoT-based student innovation projects under **TSIC**.
- Second Prize — Smart Idea Challenge (MEDHAVENSH) and Qualified MSME Idea Hackathon.
- Published article on “**Autonomous Drone Technology for Human Rescue Operations**”.
- Published Research Papers: “**Prediction of Battery Life Cycle**” (ICITSS 2024) and “**Smart Voice Assistant**” (ICEAAGE 2024).
- IBM SkillsBuild Project Management Certification and 24-Hour Hackathon participant at Medhaves BVRITH.